Equation Sample 4

Dataset-expansion sample

Derivation

The following display is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

$$
\begin{aligned}
\|f\|_{1_\mu} + \|g\|_{1_\mu} &= \int_{x \in \mathcal{D}} |f(x)| + |g(x)| \, \mathrm{d}\mu(x) \\
&= + \int_{\substack{x \in \mathcal{D} \\ f(x) \geq 0, g(x) \geq 0,}} f(x) + g(x) \, \mathrm{d}\mu(x) - \int_{\substack{x \in \mathcal{D} \\ f(x) < 0, g(x) < 0,}} f(x) + g(x) \, \mathrm{d}\mu(x) \\
&\quad + \int_{\substack{x \in \mathcal{D} \\ f(x) \geq 0, g(x) < 0,}} f(x) - g(x) \, \mathrm{d}\mu(x) + \int_{\substack{x \in \mathcal{D} \\ f(x) < 0, g(x) \geq 0,}} g(x) - f(x) \, \mathrm{d}\mu(x) \\
&= + \int_{\substack{x \in \mathcal{D} \\ f(x) \geq 0, g(x) \geq 0,}} \max(f(x), g(x)) + \min(f(x), g(x)) \, \mathrm{d}\mu(x) \\
&\quad - \int_{\substack{x \in \mathcal{D} \\ f(x) < 0, g(x) < 0,}} \max(f(x), g(x)) + \min(f(x), g(x)) \, \mathrm{d}\mu(x) \\
&\quad + \int_{\substack{x \in \mathcal{D} \\ f(x) \geq 0, g(x) < 0,}} \max(f(x), g(x)) - \min(f(x), g(x)) \, \mathrm{d}\mu(x) \\
&\quad + \int_{\substack{x \in \mathcal{D} \\ f(x) < 0, g(x) \geq 0,}} \max(f(x), g(x)) - \min(f(x), g(x)) \, \mathrm{d}\mu(x).
\end{aligned}
$$